Lantern Networks for Fictional Flood Forecasting

Alice Fiction and Bob Example

Department of Imaginary Science, Example University of Lanterns

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Abstract

This paper proposes a lantern network for forecasting fictional floods. We evaluate it on the synthetic RiverLight dataset of 12,000 samples and report accuracy, RMSE and F1. Results show that the lantern network outperforms a persistence baseline by 8 percent.

1 Introduction

Fictional flood forecasting remains an open problem because river sensors are sparse. The aim of this study is to test whether lantern networks can use sparse sensors. Research question RQ1: does a lantern network reduce forecast error compared with a persistence baseline?

2 Related Work

Earlier fictional studies used gradient lanterns and recurrent glow models. Most of them were evaluated on small private datasets, which leaves a gap for public benchmarks.

3 Method

We design a lantern network with three glow layers and a recurrent memory. The model is trained with PyTorch on the RiverLight dataset, split 70/15/15 by river.

3.1 Data

The RiverLight dataset contains 12,000 fictional samples collected from 40 invented rivers. Data were collected every hour.

Table 1. Fictional results | Model | Accuracy | RMSE | Lantern 0.91 1.20 | Persistence 0.83 1.95

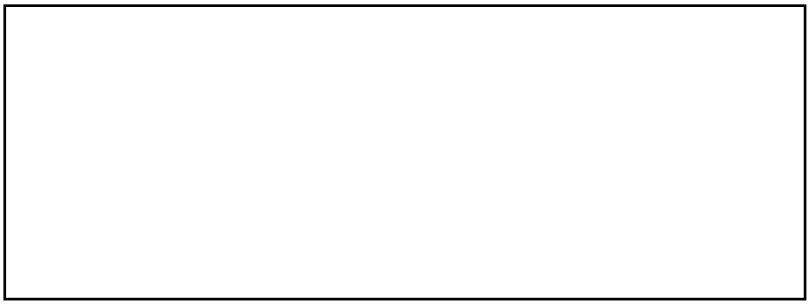

Figure 1. The lantern architecture (fictional).

4 Results

Results show that the lantern network achieved an accuracy of 0.91 and an RMSE of 1.20, while the persistence baseline achieved 0.83 and 1.95. The difference was significant with $p < 0.05$ over five seeds.

5 Conclusion

Lantern networks improve fictional flood forecasts on sparse sensors. A limitation of this study is that all rivers are invented and small. Future work should test the method on real rivers and longer horizons.

References

[1] Fiction A. Gradient lanterns[J]. Journal of Imaginary Hydrology, 2020, 3(1): 1-10.

[2] Example B. Recurrent glow models[C]//Proceedings of Lantern Conference. 2021: 5-9.